

Lab Assignment 7.4

AIAC

2303A51867

Task 1:

The screenshot shows a Google Colab notebook titled 'Untitled11.ipynb'. The code cell contains the following Python code:

```
a=4
b=4
print(a+b
```

The code has been executed, resulting in a `SyntaxError: incomplete input`. The error message is displayed in the output area. The Gemini chat interface on the right provides an explanation of the error and offers suggestions for fixing it.

Gemini

Please explain this error:

`SyntaxError: incomplete input`

The code in the selected cell has a `SyntaxError: incomplete input`. This is because the `print` function call is missing its closing parenthesis. I can fix this by adding a closing parenthesis.

The code in cell `jIXDh3CAEeF0` executed successfully and

[Print the sum of a and b](#)

[Explain the difference between printing strings and variables](#)

[Show me how to assign the sum to a new variable](#)

What can I help you build?

[+](#) Gemini 2.5 Flash [▶](#)

Gemini can make mistakes so double-check it and use code with caution. [Learn more](#)

The screenshot shows the same Google Colab notebook after the code has been corrected. The code cell now contains the following Python code:

```
a=4
b=4
print(a+b)
```

The code has been executed successfully, resulting in the output `8`. The Gemini chat interface on the right provides an explanation of the previous error and offers suggestions for fixing it.

Gemini

The code in the selected cell has a `SyntaxError: incomplete input`. This error means that the `print` function is missing its closing parenthesis. I can fix this by adding the missing parenthesis.

The code in cell `jIXDh3CAEeF0` executed successfully, and the `print(a+b)` statement outputted `8`. This is the sum of `a` (which is 4) and `b` (which is 4).

[Print the sum of a and b](#)

[Explain the difference between printing strings and variables](#)

[Show me how to assign the sum to a new variable](#)

What can I help you build?

[+](#) Gemini 2.5 Flash [▶](#)

Gemini can make mistakes so double-check it and use code with caution. [Learn more](#)

Task 2:

```
def sort_values(values):
    values.sort()
    print("Sorted list: " values)
```

File `"/tmp/ipython-input-2684529869.py"`, line 3
 print("Sorted list: " values)
 ^
SyntaxError: invalid syntax. Perhaps you forgot a comma?

Next steps: [Explain error](#)

G The following Python sorting function crashes when sorting a list containing mixed data types such as integers and strings. Analyze the code and: 1. Identify the `TypeError` cause 2. Explain why Python fails to compare mixed types 3. Highlight the faulty operation

✦ I can help analyze the issue, but first, please provide the Python sorting

[the `TypeError` for mixed data types](#) [Show an example of sorting a list with mixed data types](#)

```
def sort_values(values):
    values.sort()
    print("Sorted list:", values)
    sort_values([5, 2, 8, 1, 9])
```

Sorted list: [1, 2, 5, 8, 9]

✦ You're encountering an `IndentationError`. The lines `values.sort()` and `print('Sorted list:', values)` are indented too much. Also, the line `sort_values([5, 2, 8, 1, 9])` is mistakenly placed inside the function, which should be moved outside to call the function. I'll correct the indentation and move the function call.

[Test 'sort_values' with '\[3, 1, 4, 1, 5, 9, 2, 6\]'](#) [Add error handling to 'sort_values' for non-list i](#)

Task 3:

```
file = open("data.txt", "r"
content = file.read()
print(content)
```

File `"/tmp/ipython-input-3728278547.py"`, line 1
 file = open("data.txt", "r"
 ^
SyntaxError: '(' was never closed

Next steps: [Explain error](#)

```
with open("data.txt", "w") as f:
    f.write("This is some sample data.")
print("data.txt created successfully.")
```

data.txt created successfully.

Here's how you can create a sample 'data.txt' file with some content:

Great! The `data.txt` file has been successfully created with the content "This is some sample data.". You can now re-run the cell `JZK9k90AG9VS` to read its content.

👍 🗨

Test 'sort_values' with '[3, 1, 4, 1, 5, 9, 2, 6]' Add error handling to 'sort_values' for non-list

What can I help you build?

Task 4:

```
▶ numbers = [10, 5, 0, 2]

for n in numbers
    result = 100 / n
    print(result)
```

... File `"/tmp/ipython-input-1042031101.py"`, line 3
for n in numbers
 ^
SyntaxError: expected ':'

Next steps: [Explain error](#)

```
▶ numbers = [10, 5, 0, 2]

for n in numbers:
    try:
        result = 100 / n
        print(result)
    except ZeroDivisionError:
        print("Error: Cannot divide by zero!")
```

... 10.0
20.0
Error: Cannot divide by zero!
50.0

`except` block to handle potential `ZeroDivisionError` if any number in the list is zero.

The code executed successfully. It iterated through the `numbers` list, dividing 100 by each number. When it encountered `0`, it caught the `ZeroDivisionError` and printed "Error: Cannot divide by zero!". Otherwise, it printed the result of the division.

[Explain the try-except block in the last cell](#) [How can I handle other types of errors in this code?](#)

Task 5:

```
class Student:
    def __init__(name, roll):
        self.name = name
        roll = roll

    def display():
        print("Name:", self.name)
        print("Roll:", self.roll)

s1 = Student("Bob", 101)
s1.display()
```

... -----

TypeError Traceback (most recent call last)

/tmp/ipython-input-2809057746.py in <cell line: 0>()

8 print("Roll:", self.roll)

9

---> 10 s1 = Student("Bob", 101)

11 s1.display()

TypeError: Student.__init__() takes 2 positional arguments but 3 were given

```
class Student:
    def __init__(self, name, roll):
        self.name = name
        self.roll = roll

    def display(self):
        print("Name:", self.name)
        print("Roll:", self.roll)

s1 = Student("Bob", 101)
s1.display()
```

... Name: Bob
Roll: 101

The code in cell `cVNuo7bwI-sK` executed successfully! The `Student` object named `s1` was created with the name 'Bob' and roll number '101', and its `display()` method correctly printed:

```
Name: Bob
Roll: 101
```

[Explain "TypeError: Student.__init__\(\) takes 2 positional arguments but 3 were given"](#) [Fix the](#)